

Qatar-1b

R-1868

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_141118 · created 2026-08-03T10:24:34+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456979.61526 +/- 0.00041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.015
Transit depth (Rp/Rs)^2	2.96 +/- 0.53 [%]
Orbital Inclination (inc)	82.94 +/- 1.02
Airmass coefficient 1 (a1)	1.003 +/- 0.01
Airmass coefficient 2 (a2)	-0.002 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.01 %
Best Comparison Star	#1 - [373, 113]
Optimal Aperture	3.42
Optimal Annulus	10.45
Transit Duration (day)	0.0635 +/- 0.0076

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.172 ± 0.015 vs 0.1463 (deviation 1.71σ)
Duration fit vs published	0.0635 d vs 0.0692 d (deviation 0.75σ)
Mid-time O–C	-1.0 min
Depth SNR	11.5
Residual scatter	1.01 %

Light curve

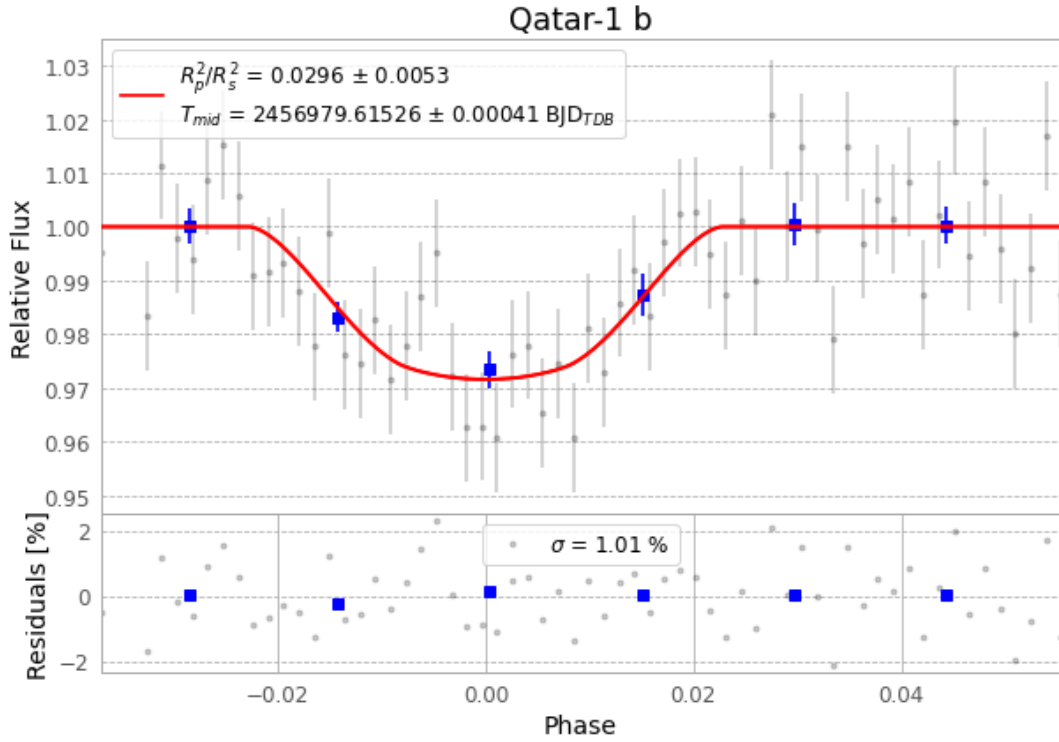


Figure 1 — FinalLightCurve_Qatar-1 b_2014-11-17.png (SHA-256 3ff3421d4a7e2515...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [396, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ab07e4870e0ff6a4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

62 FITS frames (41 MB), Qatar-1141118012713.FITS → Qatar-1141118043612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-141118.log (f8d4ee357ed9...), FinalLightCurve_Qatar-1 b_2014-11-17.png (3ff3421d4a7e...), FinalLightCurve_Qatar-1 b_2014-11-17.csv (054b8bfe9b28...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 10:24Z.